DS-973 — Data Science & Machine Learning Essentials

OFFICIAL COURSE SYLLABUS & CURRICULUM GUIDE

Course Code: DS-973	**Category:** Computer Science & Tech
Academic Term: 2026 Academic Year	**Format:** E-Learning / Self-Paced
Prerequisites: Basic Problem Solving	**Estimated Hours:** 40 Hours Content

1. Course Description & Learning Objectives

Welcome to **Data Science & Machine Learning Essentials (DS-973)**! This course is designed to equip students with deep theoretical understanding and practical hands-on proficiency. By completing this curriculum, learners will master key concepts, build industry-standard projects, and develop problem-solving skills necessary for professional software development.

Key Learning Outcomes:

• Master foundational and advanced topics through structured video lectures and interactive materials.

• Apply theoretical principles to real-world coding projects and algorithmic problem solving.

• Understand best practices, debugging strategies, and architectural design patterns.

• Pass comprehensive evaluations and earn official course certification.

2. Course Modules & Detailed Schedule

Module / Topic	Key Topics & Assignments	Weightage
Module 1: Data Science Foundations & NumPy Vectorization	In-depth lectures, reading materials, and hands-on coding exercises.	25%
Module 2: Data Cleaning & Preprocessing with Pandas	In-depth lectures, reading materials, and hands-on coding exercises.	30%
Module 3: Exploratory Data Analysis & Visualization	In-depth lectures, reading materials, and hands-on coding exercises.	35%
Module 4: Descriptive Statistics & Hypothesis Testing	In-depth lectures, reading materials, and hands-on coding exercises.	40%
Module 5: Supervised Learning & Regression Models	In-depth lectures, reading materials, and hands-on coding exercises.	45%
Module 6: Classification Algorithms & Decision Trees	In-depth lectures, reading materials, and hands-on coding exercises.	50%

3. Grading Policy & Assessment Breakdown

Component	Percentage	Details
Practical Exercises & Quizzes	30%	Weekly online quizzes & code exercises
Mid-Term Assessment	30%	Comprehensive theoretical & coding test
Final Capstone Project	40%	Individual end-to-end practical project